# KATHULAPALLI CHAKRADHAR REDDY

**Enrollment No.: BT21CSE101 • B. Tech. Computer Science and Engineering**
Visvesvaraya National Institute of Technology, Nagpur

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## EDUCATION

| Degree/Course | Institute /Board | CGPA/Percentage | Year |
|---|---|---|---|
| **B.Tech(CSE)** | Visvesvaraya National Institute of Technology, Nagpur | **7.02** | 2026 |
| CLASS XII (BIEAP) | Narayana Junior college, Vijayawada | **96.7%** | 2021 |
| CLASS X (SSC) | Dr. KKR's Gowtham International School, Vijayawada | **9.7 GPA** | 2019 |

## PROJECTS

**Cache Poisoning Mitigation using Sliding Window and Multi-Agent Reinforcement Learning (SW-MARL)** Github

- Developed a security-aware cache management framework for Information-Centric Networking (ICN/NDN) using Multi-Agent Reinforcement Learning (MARL) to prevent adversarial cache pollution attacks.
- Designed a sliding-window temporal state representation enabling routers to learn adaptive cache admission and eviction policies from dynamic traffic patterns.
- Implemented decentralized MARL agents with PyTorch and PettingZoo; evaluated hit ratio, latency, throughput, and poisoned-hit detection, achieving high attack resilience under adversarial traffic.
- **Contributed as part of a research group under Prof. Nidhi Lal; work submitted to ICICT 2026 by the team.**
  **— Tools & technologies used:** Python, PyTorch, PettingZoo, RLlib, SimPy, NetworkX, Multi-Agent Reinforcement Learning, ICN/NDN.

**Student Placement Management System** Github

- Developed a full-stack student placement platform with separate **Student**, **Company**, and **Admin** roles, enabling secure job posting, applications, and placement administration.
- Implemented **JWT-based authentication** and **role-based access control (RBAC)** to protect APIs and enforce ownership between students, companies, and applications.
- Built business rules for **duplicate applications**, **application deadlines**, **CGPA eligibility**, job ownership, and cascade deletion with centralized validation.
- Designed RESTful APIs and integrated the React frontend with the Spring Boot backend and MySQL database via Spring Data JPA.
  **— Tools & technologies used:** Java, Spring Boot, Spring Data JPA, React, MySQL, REST APIs, JWT, Maven, Git/GitHub.

**Enterprise Network Automation & Real-Time Monitoring Platform** Github

- Architected a multi-vendor automation and telemetry platform managing Cisco IOS routers & switches (Catalyst 3750, 7200) via Netmiko, FastAPI, and Ansible.
- Engineered automated Jinja2 configuration templates, Git-versioned daily backup pipelines with unified diff tracking, and drift remediation playbooks.
- Designed an automated root-cause diagnostic engine for interface errors and OSPF troubleshooting; containerized with Docker and verified via GitHub Actions CI/CD.
  **— Tools & technologies used:** Python, FastAPI, Cisco IOS, Ansible, Terraform, AWS (VPC, EC2), Docker, Netmiko, SQLite, Pytest, GitHub Actions.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, JavaScript, TypeScript, HTML, CSS, SQL.
**Frameworks & Technologies:** Spring Boot, Spring Data JPA, FastAPI, React, Node.js, REST APIs, JWT, PyTorch, PettingZoo, RLlib, SimPy, NetworkX.
**Network & DevOps:** Cisco IOS, Ansible, Docker, Kubernetes, Terraform (IaC), Netmiko, GitHub Actions, CI/CD, Linux/Bash.
**Core Skills:** Data Structures & Algorithms, OOP, Deep Reinforcement Learning, Multi-Agent Systems, Network Security.
**Databases:** MySQL, PostgreSQL, MongoDB, SQLite.
**Tools & Cloud:** VS Code, Git, GitHub, Maven, AWS (EC2, VPC, S3), Postman, GNS3, EVE-NG.

## POSITIONS OF RESPONSIBILITY

- **Event Manager**, AXIS VNIT (Central India's Largest Annual Technical Festival) • Managed event operations, logistics, technical competitions, and team coordination.
- **Alumni Secretary**, Department of Computer Science and Engineering, VNIT Nagpur • Coordinated departmental alumni relations, student interaction sessions, and career workshops.